

Learning Conditional Average Treatment Effects in Regression Discontinuity Designs using Bayesian Additive Regression Trees

Rafael Alcantara¹ P. Richard Hahn² Hedibert F. Lopes³

June 29, 2026

<https://rafaelcalcantara.github.io>

¹The University of Texas at Austin

²SoMSS, Arizona State University

³Insper Institute of Education and Research

Outline

Our contribution

Regression Discontinuity Designs (RDD)

Bayesian Additive Regression Trees (BART)

BART for causal inference

BARDDT

Simulations

Application: effect of academic probation on education

Conclusion

BARDDT Illustration

DGP details

Details on methods

Our contribution

- ▶ We propose a modification of the BART model of Chipman et al. (2010) in which the constant leaf node predictions are replaced by RDD regressions (*i.e.* regressions on the running variable and treatment dummy)
- ▶ We show that unmodified BART models estimate RDD treatment effects poorly, while our modified model accurately recovers treatment effects at the cutoff.
- ▶ At the same time, the model retains the inherent flexibility of all BART-based models, allowing it to effectively explore heterogeneous treatment effects.
- ▶ We illustrate the new method by analyzing data studied originally by Lindo et al. (2010) to estimate the effect of academic probation on university students' GPA

Leaf regression models

- ▶ Chipman et al. (2002); Gramacy and Lee (2008): linear regression/GP extensions to Bayesian CART
- ▶ Starling et al. (2019, 2020): enforce smoothness at a target covariate in the leaf predictions
- ▶ Deshpande et al. (2026): BART for varying coefficient models
- ▶ Herren et al. (2025): *stochtree* package, allows for generic basis functions for leaf regressions

Regression Discontinuity Designs

We conceptualize the treatment effect estimation problem via a quartet of random variables (Y, X, Z, U)

- ▶ Y : outcome variable;
- ▶ X : running variable;
- ▶ Z : treatment assignment indicator variable;
- ▶ U : additional, possibly unobserved, causal factors.

What specifically makes this correspond to an RDD is that we stipulate that $Z = \mathbb{I}(X > c)$, for cutoff c . We assume $c = 0$ without loss of generality.

Functional Causal Model

We may express Y as some function of the random variables (X, Z, U) :

$$Y = F(X, Z, U).$$

This formulation relates to the potential outcomes framework straightforwardly:

$$\begin{aligned} Y^1 &= F(X, 1, U), \\ Y^0 &= F(X, 0, U). \end{aligned} \tag{1}$$

Next, we define the following conditional expectations

$$\begin{aligned}\mu_1(x) &= \mathbb{E}[F(x, 1, U) \mid X = x], \\ \mu_0(x) &= \mathbb{E}[F(x, 0, U) \mid X = x],\end{aligned}\tag{2}$$

with which we can define the treatment effect function

$$\tau(x) = \mu_1(x) - \mu_0(x).$$

Identification

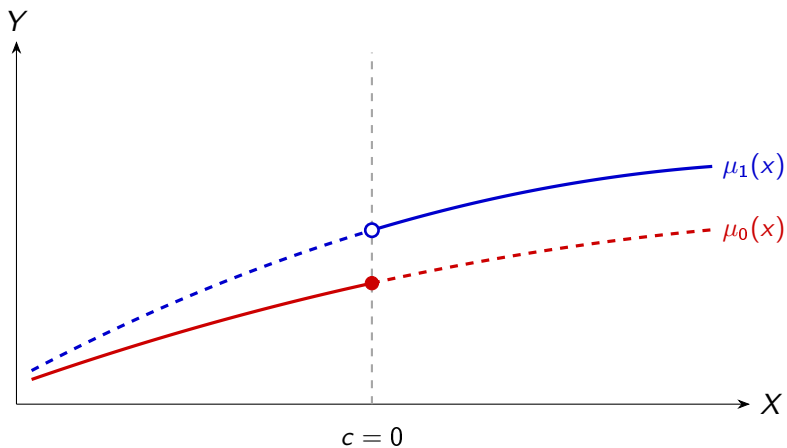
Because of the deterministic treatment assignment rule, we can learn $\mu_1(x)$ for $X > 0$ and $\mu_0(x)$ for $X < 0$:

$$\begin{aligned}\mu_1(x) &= \mathbb{E}[F(x, 1, U) \mid X = x] = \mathbb{E}[Y \mid X = x, Z = 1] \quad \text{for } x > 0, \\ \mu_0(x) &= \mathbb{E}[F(x, 0, U) \mid X = x] = \mathbb{E}[Y \mid X = x, Z = 0] \quad \text{for } x \leq 0\end{aligned}$$

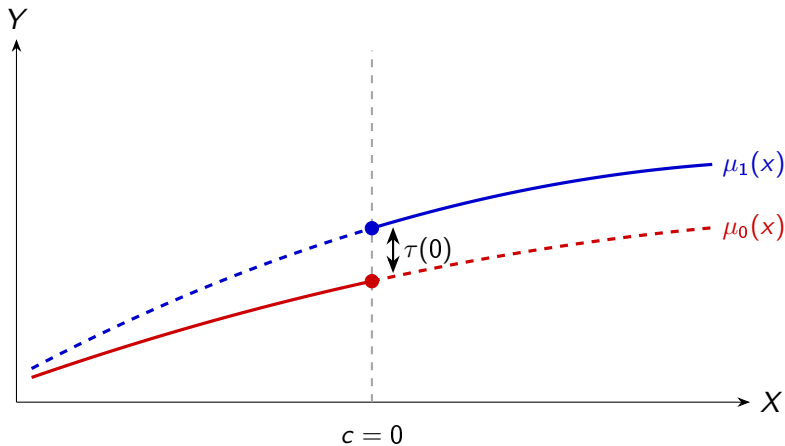
Identification assumption: If $\mu(x)$ is continuous at $x = 0$, it is possible to learn $\tau(0)$ as the difference between $\mu_1(0) - \mu_0(0)$

- ▶ Any inferred discontinuity at $x = 0$ must be attributable to treatment effect (Hahn et al., 2001)

An illustration



An illustration



Re-define our potential outcome means as

$$\begin{aligned}\mu_1(x, w) &= \mathbb{E}[F(x, 1, U) \mid X = x, W = w] \\ \mu_0(x, w) &= \mathbb{E}[F(x, 0, U) \mid X = x, W = w],\end{aligned}\tag{3}$$

and our treatment effect function as

$$\tau(x, w) = \mu_1(x, w) - \mu_0(x, w).$$

Assumptions:

- ▶ $\mu_1(x, w)$ and $\mu_0(x, w)$ are continuous at $x = 0$, for every w
- ▶ $0 < P(W = w \mid X = x) < 1$ for every pair (w, x)

Brief BART review

Letting $f(x) = E(Y \mid X = x)$ denote a smooth function of a covariate vector X , the BART model is traditionally written

$$\begin{aligned} Y_i &= f(x_i) + \varepsilon_i \\ &= \sum_{j=1}^k g_j(x_i; T_j, m_j) + \varepsilon_i \end{aligned} \tag{4}$$

where $\varepsilon_i \sim N(0, \sigma^2)$ is a normally distributed additive error term

$g_j(x; T_j, m_j)$: piecewise function of x defined by a set of splitting rules T_j that partitions the domain of x into disjoint regions, and a vector, m_j , which records the values $g(\cdot)$ takes on each of those regions

BART for causal inference

S-learners: BART with treatment as covariate (Hill, 2011).

T-learners: Two BART models (Künzel et al., 2019).

These approaches are not ideal in common causal inference settings (Hahn et al., 2020):

T-learner: regularization of the treatment effect is necessarily weaker than regularization of each individual model.

S-learner: joint distribution of the control variables and the treatment variable may lead to unpredictable biases.

BARDDT

We propose a BART model where the trees are allowed to split on (x, w) but where each leaf node parameter is a vector of regression coefficients tailored to the RDD context

Basis vector

$$\psi(x, z) = [1 \quad zx \quad (1 - z)x \quad z] \quad (5)$$

Prediction for point (x, w, z) at node b_j of tree j

$$g_j(x, w, z) = \psi(x, z) \Gamma_{b_j(x, w)} \quad (6)$$

Posterior sampling from this model proceeds nearly identically to the traditional BART Gibbs sampler

- ▶ Modified log marginal likelihood, which generalizes the one from constant-leaf BART to one from a node-level regression with the basis defined above
- ▶ Parameter sampling follows a standard conditionally (on σ^2) conjugate linear regression update, independently for each leaf of the current tree (see for example, section 2.3.3 in Gamerman and Lopes (2006))

Simulations - Estimand

Generically, our estimand is $\tau(0, w)$. To focus on feasible points, we recommend restricting the evaluation points to the observed w_i such that $|x_i| \leq \delta$, for some $\delta > 0$. Therefore, our estimand of interest is a vector of treatment effects:

$$\tau(0, w_i) \quad \forall i \text{ such that } |x_i| \leq \delta. \quad (7)$$

In our example, we use $\delta = 0.1$ for a standardized x variable.

Simulations - Estimation loss function

For our evaluation criteria we will consider average root-mean-squared estimation error, expressed as a fraction of a default ATE estimator:

$$\text{CATE RMSE} = \frac{\sqrt{\sum_{i:|x_i| \leq \delta} (\hat{\tau}(0, w_i) - \tau(0, w_i))^2}}{\sqrt{\sum_{i:|x_i| \leq \delta} (\hat{\tau}(0) - \tau(0, w_i))^2}}. \quad (8)$$

This performance metric judges the ability of $\hat{\tau}(0, w)$ to estimate CATEs relative to a baseline ATE estimator (at $x = 0$)

- ▶ Check if methods are doing better than would be possible just by assuming homogeneous effects
- ▶ Relative accuracy can be compared in a standardized way across data generating processes of varying outcome scales

We design our DGP so that it is explicitly parametrized in terms of conditional variances at $x = 0$

DGP which can be roughly separated into two groups: “easy” and “hard”

Easy setting: prognostic and treatment variation are comparable magnitudes, μ is separable in x and w , and low noise

Hard setting: prognostic variation is twenty times larger than treatment variation, μ is non-separable in x and w , and noise is high

Results are based on 100 replications of size $n = 4000$ for each DGP configuration.

To demonstrate our simulation protocol we will compare the following methods:

- ▶ BARDDT
- ▶ S-BART
- ▶ T-BART
- ▶ a local polynomial estimator (with and without regularization)
- ▶ RD-Tree (Reguly, 2021).

The CATE estimator in all cases was the vector of posterior means of $\tau(0, w_i)$ for i such that $|x_i| \leq 0.1$.

[Details on methods](#)

Summary of Results - BART alternatives

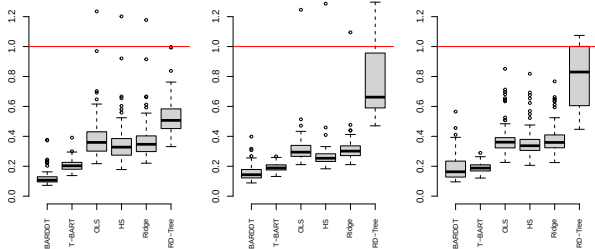
- ▶ T-BART's high variance results in substantially higher RMSE relative to both BARDDT and the polynomial model.
- ▶ Extreme bias shift exhibited by S-BART in the low noise setting is reminiscent of the regularization-induced confounding (RIC) problem, described by Hahn et al. (2020).
- ▶ S-BART does comparatively well in the high noise case, but only because it rarely splits in that case, collapsing to a homogenous treatment model, which outperforms the overfitting T-BART and polynomial models in this regime.

Summary of Results - Other estimators

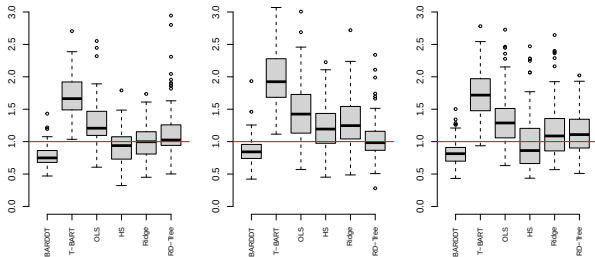
- ▶ Even with high signal, the polynomial models struggles with extrapolation at the boundaries of the support of w_1 .
- ▶ Polynomials model also presents a sizable increase in variance under high noise. Regularization decreases the variance significantly, but the model still suffers from misspecification
- ▶ RD-Tree appears to “under-split” on W , leading to a too-coarse fit of the CATE function, especially in the low-noise regime.

Overall Results

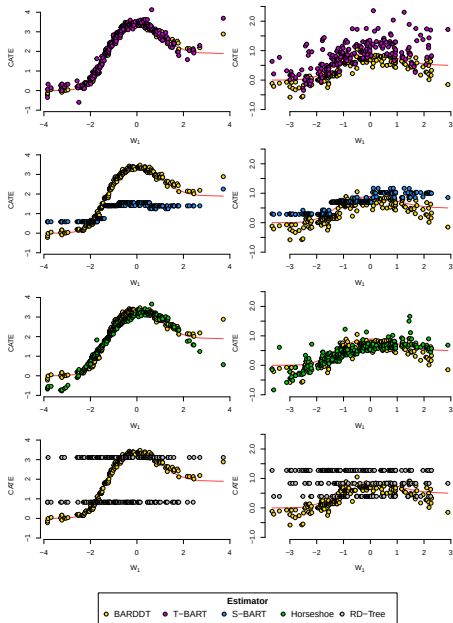
High signal, separable μ



Low signal, non-separable μ



Individual Fits



Application: effect of academic probation on education

- ▶ We investigate the effect of academic probation in educational outcomes in a large Canadian university (Lindo et al., 2010)
- ▶ Students who, by the end of each term, present GPA lower than a certain threshold (which differs between each campus) are placed on academic probation and must improve their GPA in the next term
- ▶ Punishment if they fail to achieve this goal can range from 1-year to permanent suspension from the university
- ▶ We focus on GPA in the term after a student is placed on probation

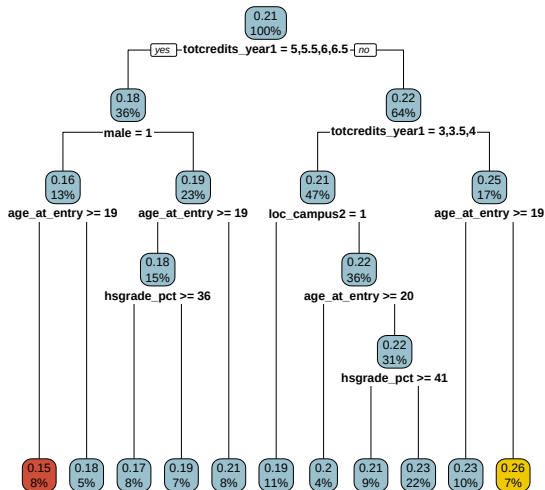
Application

- ▶ Running variable is the (negative) distance between a student's GPA and the probation threshold, meaning students below the limit have a positive score and the cutoff is 0
- ▶ Additional student features: gender, age, a *dummy* for being born in North America, attempted credits in the first year, *dummies* for which campus each student belongs to, and the student's position in the distribution of high school grades of students entering the university in the same year as a measure of high school performance.

Application: fit-the-fit

- ▶ As in Hahn et al. (2020), we explore the individual effect estimates – the posterior mean of the individual effects – by fitting a CART tree to these estimates based on the covariate set ('fit-the-fit')
- ▶ With this strategy, we allow the data to determine relevant treatment effective modifiers and potential interactions between them

Application: fit-the-fit



Application: fit-the-fit

Consider the two groups of students at opposite ends of the treatment effect range discovered by the effect moderation tree:

Group A a male student that entered college older than 19 and attempted at least 5 credits in the first year (leftmost leaf node, colored red, comprising 128 individuals)

Group B a student of any gender who entered college younger than 19 and attempted more than 4, but less than 5 credits in the first year (rightmost leaf node, colored gold, comprising 108 individuals).

Application: fit-the-fit

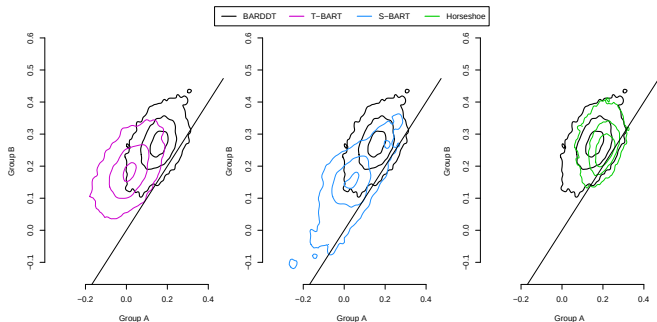
Subgroup CATEs are obtained by aggregating CATEs across the observed w_i values for individuals in each group; this can be done for individual posterior samples, yielding a posterior distribution over the subgroup CATE:

$$\bar{\tau}_A^{(h)} = \frac{1}{n_A} \sum_{i:w_i} \tau^{(h)}(0, w_i), \quad (9)$$

where h indexes a posterior draw and n_A denotes the number of individuals in the group A.

Application - Posterior Comparisons I

Figure 1: Kernel density estimates for the joint CATE posterior between male students who entered college older than 19 and attempted at least 5 credits in the first year (leftmost leaf node, red) and students who entered college younger than 19 and attempted between 4 and 5 credits in the first year (rightmost leaf node, gold). Black contour lines show the joint CATE posterior obtained by BARDDT, which is contrasted with the joint posterior obtained by T-BART, S-BART and horseshoe, respectively



Conclusion

- ▶ **Main contributions:** incorporating RDD assumptions into the BART framework and producing reliable CATE estimates
- ▶ **Results:** BARDDT presents lower errors and more stable performance compared to other commonly used estimators based on parametric specifications, and to off-the-shelf BART implementations
- ▶ **Extensions:** extrapolating the estimates beyond the cutoff (Wang et al., 2023), modelling non-Gaussian outcomes — e.g. discrete or t-distributed — and extending our strategy for settings with multiple cutoffs

Thank you!

Personal website: `rafaelcalcantara.github.io`

Github repo: `https://github.com/rafaelcalcantara/BART-RDD`

Final references I

- Calonico, S., Cattaneo, M. D., Farrell, M. H., Palomba, F., and Titiunik, R. (2025). Treatment effect heterogeneity in regression discontinuity designs. *arXiv preprint arXiv:2503.13696*.
- Calonico, S., Cattaneo, M. D., and Titiunik, R. (2015). rdrobust: An r package for robust nonparametric inference in regression-discontinuity designs. *R J.*, 7(1):38.
- Carvalho, C. M., Polson, N. G., and Scott, J. G. (2009). Handling sparsity via the horseshoe. In *Artificial intelligence and statistics*, pages 73–80. PMLR.
- Chipman, H. A., George, E. I., and McCulloch, R. E. (2002). Bayesian treed models. *Machine Learning*, 48:299–320.
- Chipman, H. A., George, E. I., McCulloch, R. E., et al. (2010). Bart: Bayesian additive regression trees. *The Annals of Applied Statistics*, 4(1):266–298.

Final references II

- Deshpande, S. K., Bai, R., Balocchi, C., Starling, J. E., and Weiss, J. (2026). Vcbart: Bayesian trees for varying coefficients. *Bayesian Analysis*, 21(1):281–308.
- Gamerman, D. and Lopes, H. F. (2006). *Markov chain Monte Carlo: stochastic simulation for Bayesian inference*. Chapman and Hall/CRC.
- Gramacy, R. B. and Lee, H. K. H. (2008). Bayesian treed Gaussian process models with an application to computer modeling. *Journal of the American Statistical Association*, 103(483):1119–1130.
- Hahn, J., Todd, P., and Van der Klaauw, W. (2001). Identification and estimation of treatment effects with a regression-discontinuity design. *Econometrica*, 69(1):201–209.

Final references III

- Hahn, P. R., Murray, J. S., Carvalho, C. M., et al. (2020). Bayesian regression tree models for causal inference: regularization, confounding, and heterogeneous effects. *Bayesian Analysis*.
- Herren, A., Hahn, P. R., Murray, J., and Carvalho, C. (2025). Stochtree: Bart-based modeling in r and python. *arXiv preprint arXiv:2512.12051*.
- Hill, J. L. (2011). Bayesian nonparametric modeling for causal inference. *Journal of Computational and Graphical Statistics*, 20(1):217–240.
- Künzel, S. R., Sekhon, J. S., Bickel, P. J., and Yu, B. (2019). Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the national academy of sciences*, 116(10):4156–4165.

Final references IV

- Lindo, J. M., Sanders, N. J., and Oreopoulos, P. (2010). Ability, gender, and performance standards: Evidence from academic probation. *American Economic Journal: Applied Economics*, 2(2):95–117.
- Reguly, A. (2021). Heterogeneous treatment effects in regression discontinuity designs. *arXiv preprint arXiv:2106.11640*.
- Starling, J. E., Murray, J. S., Carvalho, C. M., Bukowski, R. K., Scott, J. G., et al. (2020). Bart with targeted smoothing: An analysis of patient-specific stillbirth risk. *Annals of Applied Statistics*, 14(1):28–50.
- Starling, J. E., Murray, J. S., Lohr, P. A., Aiken, A. R., Carvalho, C. M., and Scott, J. G. (2019). Targeted smooth bayesian causal forests: An analysis of heterogeneous treatment effects for simultaneous versus interval medical abortion regimens over gestation. *arXiv preprint arXiv:1905.09405*.

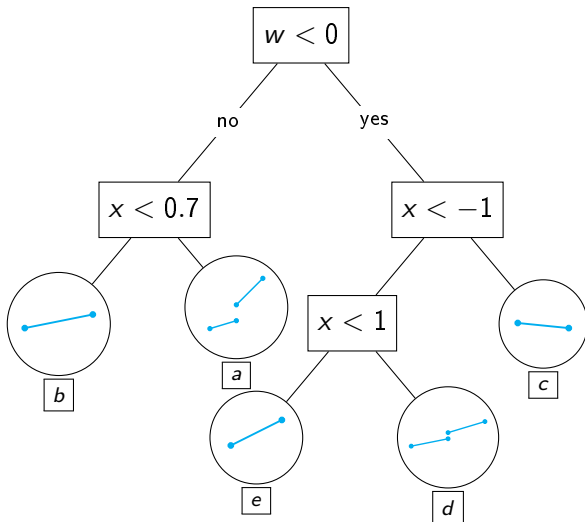
Wang, M., He, J., and Hahn, P. R. (2023). Local gaussian process extrapolation for bart models with applications to causal inference. *Journal of Computational and Graphical Statistics*, (just-accepted):1–22.

Section 9

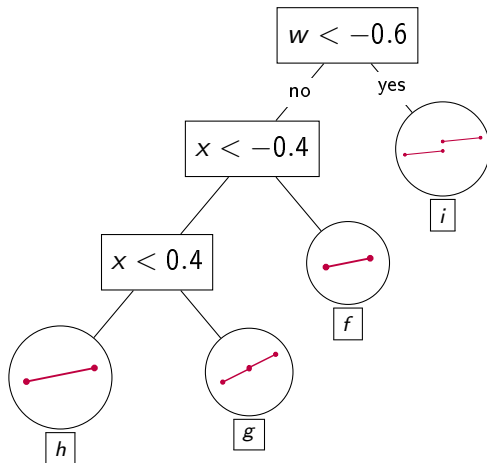
BARDDT Illustration

Main slides

BARDDT-Illustration I



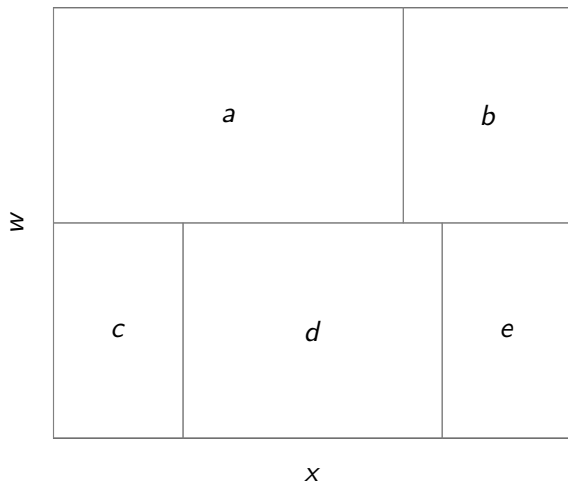
BARDDT-Illustration II



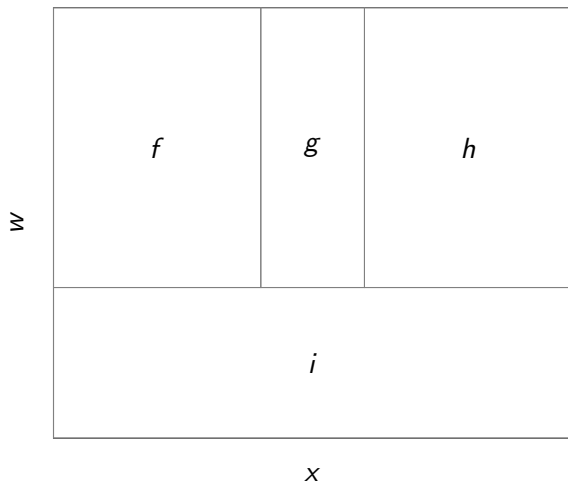
BARDDT-Illustration III

Figure 2: Two regression trees with splits in x and a single scalar w . Node images depict the $g(x, w, z)$ function (in x) defined by that node's Γ coefficients. The vertical gap between the two line segments in a node that contain $x = 0$ is that node's contribution to the CATE at $X = 0$. Note that only such nodes contribute for CATE prediction at $x = 0$

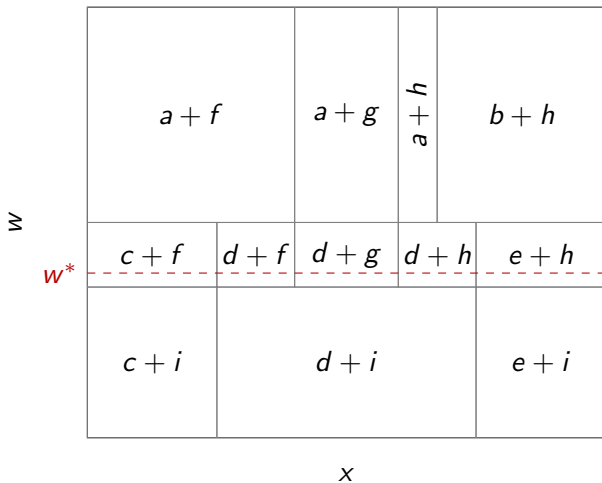
BARDDT-Illustration IV



BARDDT-Illustration V



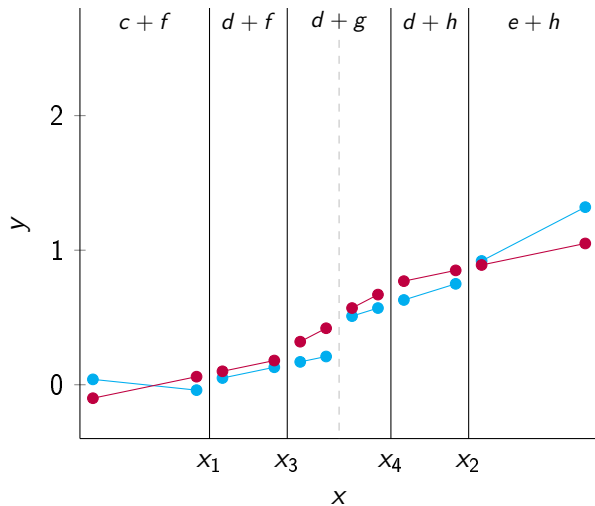
BARDDT-Illustration VI



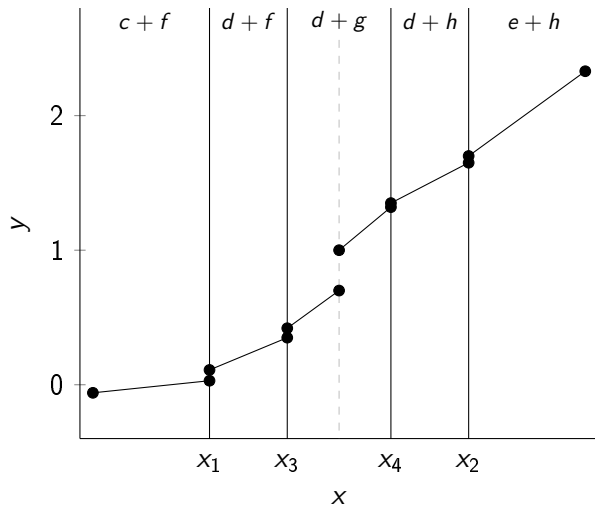
BARDDT-Illustration VII

Figure 3: The two top figures show the same two regression trees as in the preceding figure, now represented as a partition of the x - w plane. Labels in each partition correspond to the leaf nodes depicted in the previous picture. The bottom figure shows the partition of the x - w plane implied by the sum of the two trees; the red dashed line marks point $W = w^*$ and the combination of nodes that include this point

BARDDT-Illustration VIII



BARDDT-Illustration IX



BARDDT-Illustration X

Figure 4: Left: The function fit at $W = w^*$ for the two trees shown in the previous two figures, shown superimposed. Right: The aggregated fit achieved by summing the contributes of two regression tree fits shown at left. The magnitude of the discontinuity at $x = 0$ (located at the dashed gray vertical line) represents the treatment effect at that point. Different values of w will produce distinct fits; for the two trees shown, there can be three distinct fits based on the value of w .

Section 10

DGP details

Main slides

Generating (W, X)

Our simulation studies will consider W to be fixed in advance and we will consider replications over (X, Y)

We generate W according to a mean-zero multivariate Gaussian distribution with a Toeplitz covariance matrix, with entries ranging from 0 to 2

We then draw X according to a Gaussian distribution centered at a linear combination of the $W = w$ values:

$$X \mid W = w \sim N(\gamma_0 + w^t \gamma, \nu)$$

where γ_0 is the marginal mean and γ is a p -dimensional vector of regression coefficients

Generating Y given (W, X)

We generate Y according to:

$$Y_i = \mu(x_i, w_i) + \tau(x_i, w_i)z_i + \sigma\epsilon_i, \quad (10)$$

for a mean-zero Gaussian error term

We consider the following quantities: $\min_w \tau(0, w)$, $\mathbb{V}(\tau(0, W) \mid X = 0)$, and $\mathbb{V}(\mu(0, W) \mid X = 0)$

We take a large sample from $W \mid X = 0$ and compute the above quantities based on that simulated data. For (W, X) draw as described above, $W \mid X = 0$ is a multivariate Gaussian with

$$\begin{aligned} \mathbb{E}(W \mid X = 0) &= -\gamma_0\gamma, \\ \mathbb{V}(W \mid X = 0) &= \Sigma_W - \Sigma_W\beta\beta^t\Sigma_W^t. \end{aligned} \quad (11)$$

Generating Y given (W, X)

We fix $\mathbb{V}(\mu(0, W) \mid X = 0) = 1$ and specify our DGP in terms of the following parameters

$$\begin{aligned}\sqrt{\mathbb{V}(\tau(0, W) \mid X = 0)} &= k_2, \\ \sqrt{\mathbb{V}(Y \mid X = 0, W)} &= \sigma = k_4 \\ \min_w \tau(0, w) &= \underline{\tau}_0 = k_5.\end{aligned}\tag{12}$$

Finally, letting $w = (w_1 \dots w_p)$ be realizations of a length p random vector W , define $w^* = \frac{\sum_{j=1}^p w_j}{\sqrt{p}}$. Our template functions are:

$$\begin{aligned}\mu^*(x, w) &= k_1(x+1)^3 + (w^* + 2)^2 \left(\text{sign}(x+1) \sqrt{|(x+1)|} \right)^{k_3}, \\ \tau^*(w) &= \Phi(2w_1 + 3)/2 + \phi(w_1),\end{aligned}\tag{13}$$

where $\Phi(\cdot)$ and $\phi(\cdot)$ are the cumulative distribution and probability density functions, respectively, of a standard normal random variable

Generating Y given (W, X)

Parameter k_1 controls how much variability of μ will be due to X versus to W and k_3 determines whether or not μ is additive in X and W or if there is an interaction ($k_3 = 1$) or not ($k_3 = 0$)

Section 11

Details on methods

Main slides

Methods

All three BART variants were fit with 50 trees each (two forests of 50 trees for T-BART), with tree depth parameters set as in Chipman et al. (2010): $\alpha = 0.95, \beta = 2$ and fit using the `stochtree` package.

The local polynomial estimator is trained on data points within the bandwidth obtained with the `rdrobust` package (Calonico et al., 2015). Ordinary-least-squares is used to fit a fourth degree polynomial in each feature of W , interacted with X and Z :

$$Y \sim \left(\sum_{j=1}^P \text{poly}(W_j, 4) \right) \cdot X \cdot Z, \quad (14)$$

where $\text{poly}(A, p) = \sum_{j=1}^p A^j$. This choice of a fully saturated model that is linear in X is consistent with the suggested regression in Calonico et al. (2025). For the regularized model, we considered the horseshoe prior Carvalho et al. (2009) and the ridge prior

Model parameters for RD-Tree were set as suggested in Reguly (2021)